Lesson 27

30.03.2025

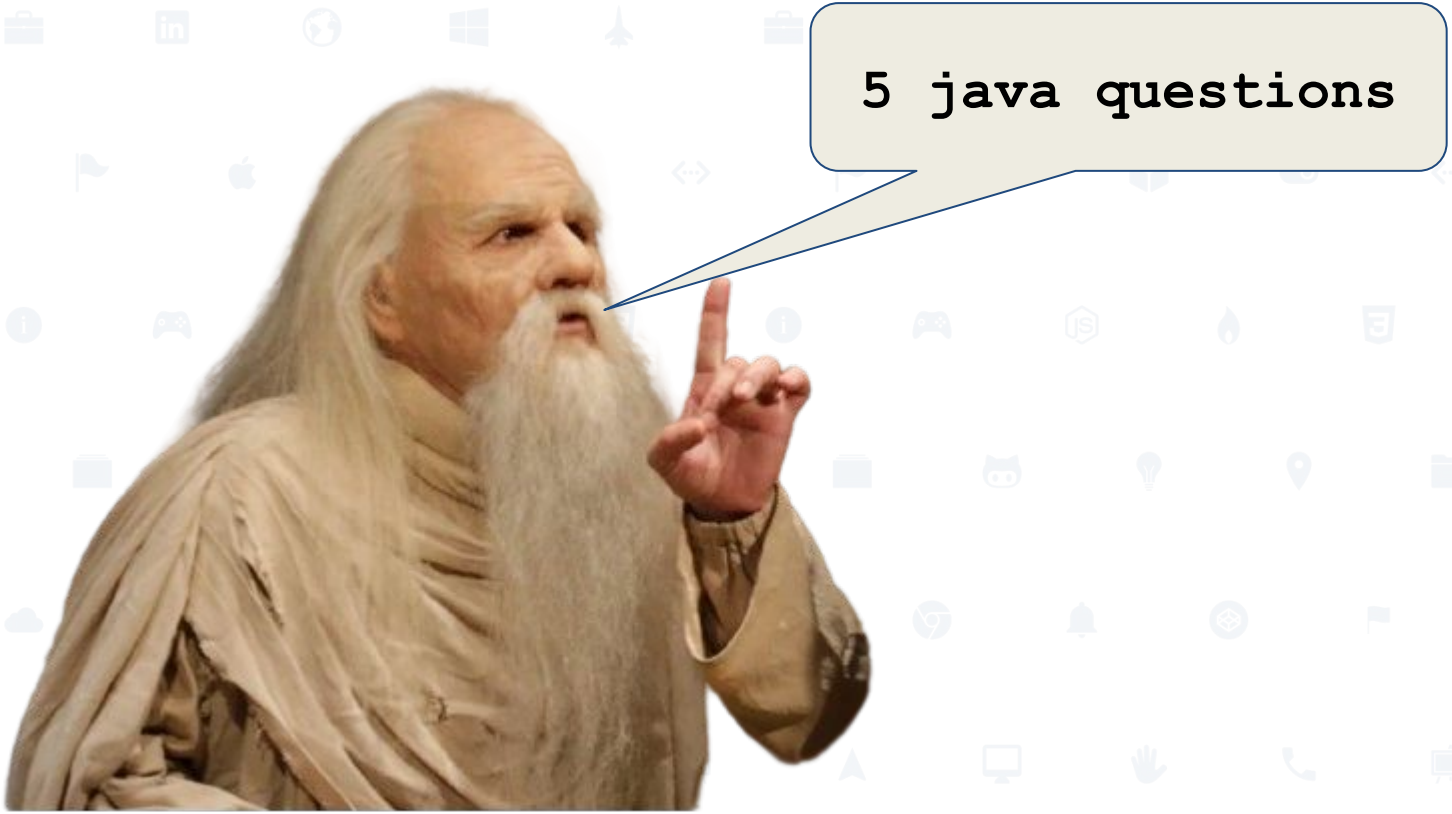
```
public class Test1 {
    public static void main(String[] args) {
        int sum = 0;
        for (int i = 0, j = 0; i < 6 & j < 5; ++i, j = i + 1) {
            sum = +i;
            System.out.println(sum);
        }
    }
}
```

```java
public class Test2 {
    public static void main(String[] args) {
        int arr[] = {11, 22, 33};
        for (int i = 0; i < arr.length; i++)
            System.out.println(arr[i] + " ");

        int arrr[] = new int[3];
        arrr[] = {11, 22, 33};
        for (int i = 0; i < arrr.length; i++)
            System.out.println(arrr[i] + " ");
    }
}
```

```java
public class Test3 {
    public static void main(String[] args) {
        int[][] arr1 = new int[2][3];
        int[][] arr2 = new int[2][];
        int[][] arr3 = new int[][];
        int[][] arr4 = new int[][3];
    }
}
```

```java
public class Test4 {
    public static void main(String[] args) {
        for (int i = 0; i < 1; System.out.println("Java")) {
            System.out.println("Scala");
        }
    }
}
```


5 java questions

1. Чим відрізняється == від методу .equals() у Java?
2. Що таке **автобоксинг** і **розпакування** (*autoboxing/unboxing*)?
3. Що відбудеться, якщо спробувати змінити об’єкт типу String?
4. Що таке final, finally і finalize() — у чому відмінності?
5. Що таке ArrayList і чим він відрізняється від LinkedList?

```sql
ALTER TABLE table_name
ADD column_name datatype;
```

```sql
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

```sql
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

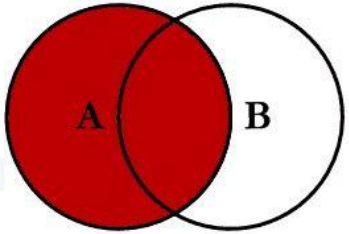
```sql
SELECT column1, column2, ...
FROM table_name;
```

```sql
SELECT COUNT(column_name)
FROM table_name
WHERE condition;
```

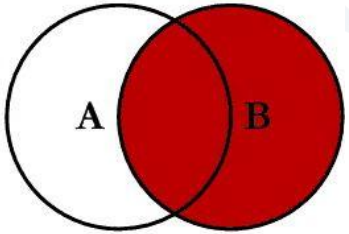
```sql
SELECT AVG(column_name)
FROM table_name
WHERE condition;
```

```sql
SELECT SUM(column_name)
FROM table_name
WHERE condition;
```

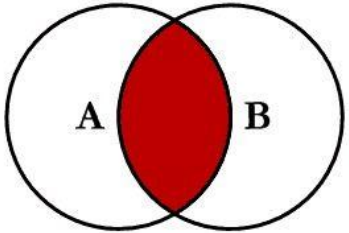
SQL JOINS



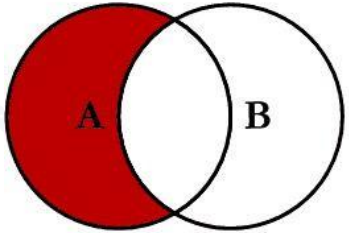

```
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
```



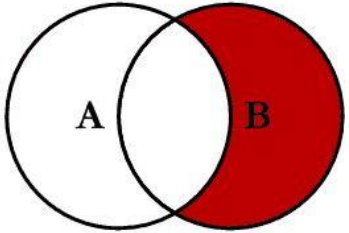

```
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
```



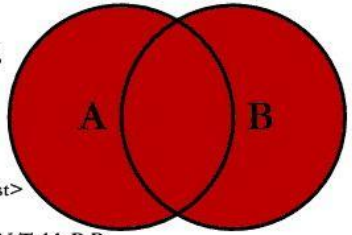

```
SELECT <select_list>
FROM TableA A
INNER JOIN TableB B
ON A.Key = B.Key
```



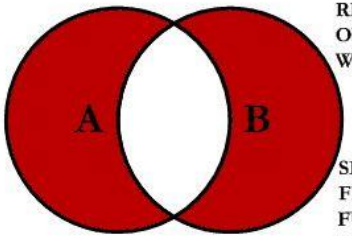

```
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
WHERE B.Key IS NULL
```




```
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
```



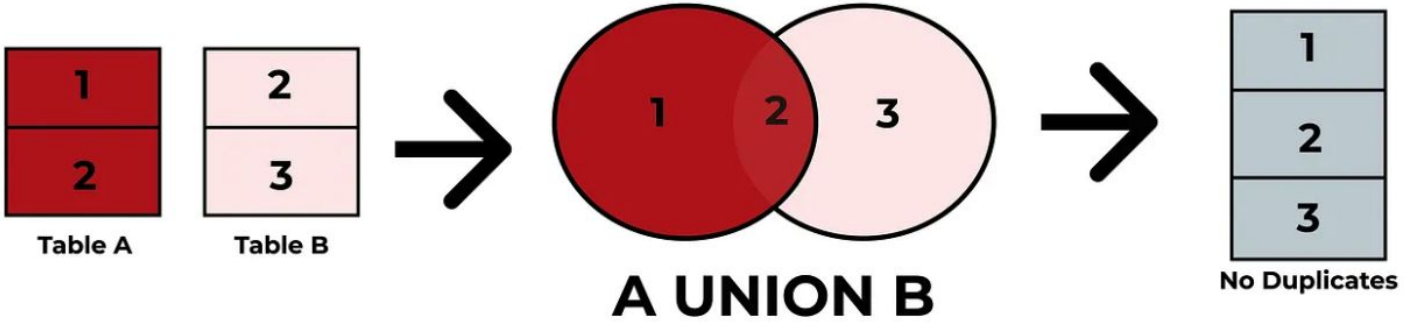

```
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
```

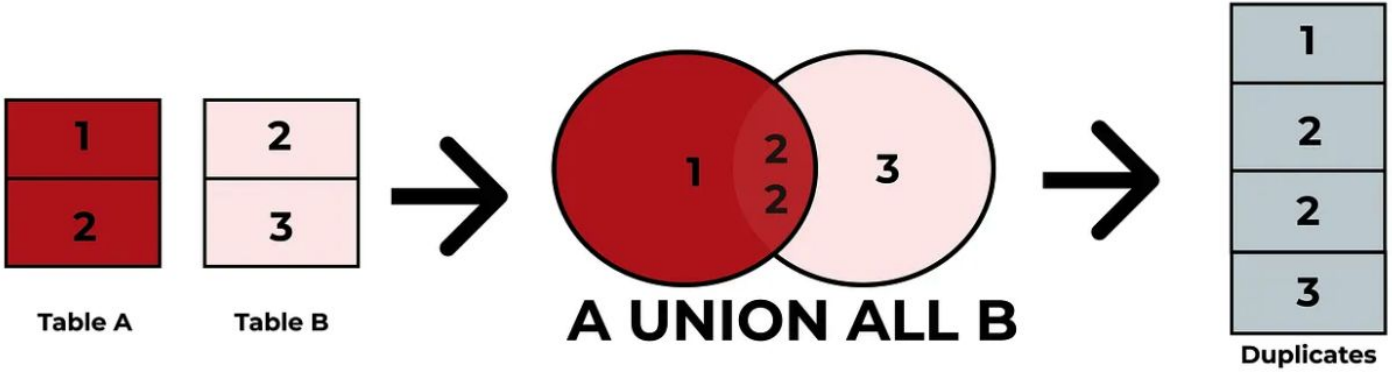



```
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
OR B.Key IS NULL
```

© C.L. Moffatt, 2008

SQL UNIONs


1
2
Table A
2
3
Table B
1
2
3
A UNION B
1
2
3
No Duplicates


1
2
Table A
2
3
Table B
1
2
2
3
A UNION ALL B
1
2
2
3
Duplicates

Employee

EmployeeID	Ename	DeptID	Salary
1001	John	2	4000
1002	Anna	1	3500
1003	James	1	2500
1004	David	2	5000
1005	Mark	2	3000
1006	Steve	3	4500
1007	Alice	3	3500

SELECT *DeptID, AVG(Salary)*
FROM *Employee*
GROUP BY *DeptID;*

GROUP BY Employee Table using DeptID

DeptID	AVG(Salary)
1	3000.00
2	4000.00
3	4250.00

Aggregate Functions

SUM(): Returns the sum or total of each group.
COUNT(): Returns the number of rows of each group.
AVG(): Returns the average and mean of each group.
MIN(): Returns the minimum value of each group.
MAX(): Returns the minimum value of each group.

Employee

EmployeeID	Ename	DeptID	Salary
1001	John	2	4000
1002	Anna	1	3500
1003	James	1	2500
1004	David	2	5000
1005	Mark	2	3000
1006	Steve	3	4500
1007	Alice	3	3500

SELECT *DeptID, AVG(Salary)*
FROM *Employee*
GROUP BY *DeptID;*

GROUP BY Employee Table using DeptID

DeptID	AVG(Salary)
1	3000.00
2	4000.00
3	4250.00

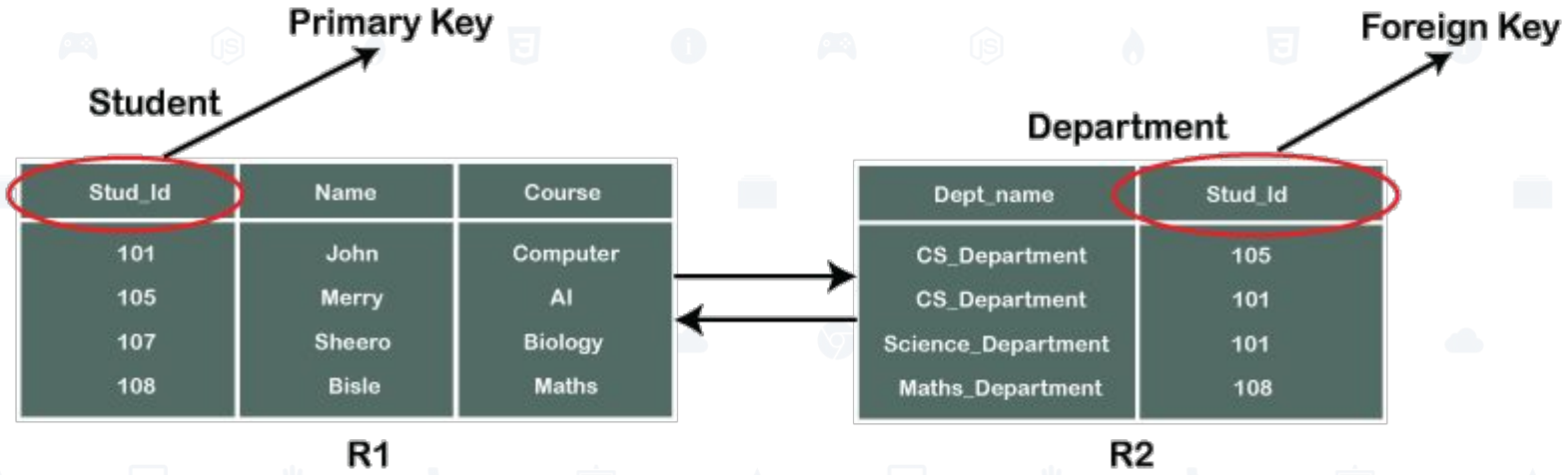
SELECT *DeptID, AVG(Salary)*
FROM *Employee*
GROUP BY *DeptID*
HAVING *AVG(Salary) > 3000;*

HAVING

DeptID	AVG(Salary)
2	4000.00
3	4250.00

Primary key (PK)

У кожній таблиці БД може бути первинний ключ. Під первинним ключем розуміють поле або набір полів, що однозначно (унікально) ідентифікують запис. Первинний ключ має бути мінімально достатнім: він має складатися з полів, видалення яких із первинного ключа не позначиться з його унікальності.

Foreign key(FK)

Забезпечує однозначний логічний зв'язок між таблицями однієї БД.